Decomposition Strategy Benchmarks and Complexity Analysis for Quantum-Classical Hybrid Food Production Optimization

Automated Benchmark Report

March 12, 2026

Abstract

We benchmark the decomposition strategies used in Study 2.A (Binary Patch allocation, Variant A) and Study 2.B (Crop Rotation, Variant B) of the UC-002 food production optimization framework, measuring their computational overhead and scaling behaviour across problem sizes from 5 to 10000 farms (with runtime caps for the heaviest Variant B cases). We compare three solving paradigms: (i) Gurobi solving the monolithic problem, (ii) Gurobi solving independently decomposed sub-problems followed by solution merging, and (iii) Parallel Tempering with Isoenergetic Cluster Moves (PT-ICM) on the decomposed sub-problems. Finally, we analyse the reducibility of both problem formulations to MAX-k-SAT.

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1 Problem Formulations

1.1 Variant A — Binary Patch Allocation (Study 2.A)

Variant A assigns one of 27 crops to each of F farms. Let $Y_{f,c} \in \{0,1\}$ indicate whether farm f grows crop c, and $U_c \in \{0,1\}$ indicate whether crop c is grown anywhere.

Objective.

$$\max \sum_{f=1}^{F} \sum_{c=1}^{C} b_c \frac{a_f}{A_{\text{tot}}} Y_{f,c} \tag{1}$$

where b_c is a composite benefit score (nutritional value, sustainability, affordability, etc.) and a_f/A_{tot} is the area fraction of farm f.

Constraints.

$$\sum_{c=1}^{C} Y_{f,c} \leq 1 \qquad \forall f \tag{2}$$

$$U_c \geq Y_{f,c} \qquad \forall f, c \tag{3}$$

$$\sum_{c \in G_g} U_c \geq 2 \qquad \forall \text{food groups } g \tag{4}$$

The decision space has $n = FC + C$ binary variables (F farms $\times$ C crops plus C usage indicators).

1.2 Variant B — Crop Rotation (Study 2.B)

Variant B assigns one of 27 crops to each farm across $T = 3$ time periods. Let $Y_{f,c,t} \in \{0,1\}$ indicate farm f grows crop c in period t.

QUBO objective.

$$\min \sum_{f,k,t} h_{f,k,t} Y_{f,k,t} + \sum_{\substack{(f,k,t) \\ (f',k',t')}} J_{(f,k,t),(f',k',t')} Y_{f,k,t} Y_{f',k',t'} \tag{5}$$

where the linear terms encode crop benefit scores, and the quadratic terms encode:

- **Temporal rotation synergies:** $R_{c,c'}$ between consecutive periods on the same farm,
- **One-hot penalties:** enforcing exactly one crop per farm per period.

The decision space has $n = 27FT = 81F$ binary variables.

2 Decomposition Strategies

2.1 Variant A Decompositions

PlotBased: One sub-problem per farm plus one for the global U variables. Partition count: $F + 1$; max sub-problem: C variables. Complexity: $O(FC)$.

Multilevel(g): Group farms into clusters of g. Partition count: $\lceil F/g \rceil + 1$; max sub-problem: gC. Complexity: $O(FC)$.

Louvain: Constructs a variable interaction graph and applies Louvain community detection (resolution 1.5). Partition count varies; complexity dominated by graph construction $O(FC^2)$ and Louvain $O(m)$ where $m = FC^2/2 + FC$.

Spectral(k): Spectral clustering on the adjacency matrix of the variable interaction graph. Requires $O(n^2)$ memory and $O(n^2k)$ time for eigendecomposition, where $n = FC + C$.

HybridGrid(f_g, c_g): 2-D grid over farms (step f_g) $\times$ crops (step c_g) plus a global U partition. Partition count: $\lceil F/f_g \rceil \cdot \lceil C/c_g \rceil + 1$.

Coordinated / CQM-First: Same partition structure as PlotBased; differs only in the solving pipeline (master–subproblem coordination vs. CQM-first).

2.2 Variant B Decompositions

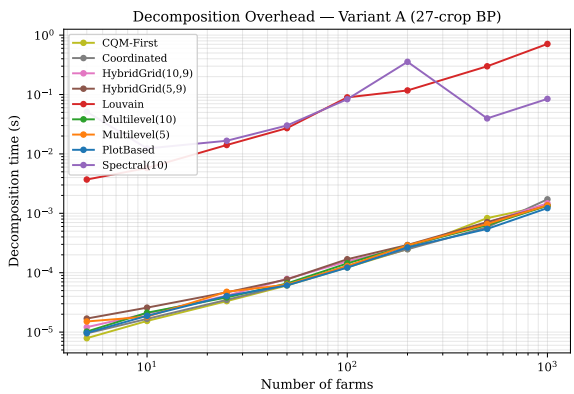
Clique (farm-by-farm): One sub-problem per farm, each with $27T = 81$ variables. Partition count: F. Ignores inter-farm spatial couplings.

SpatialTemporal(g): Cluster g farms $\times$ one period per partition. Partition count: $\lceil F/g \rceil \cdot T$; max sub-problem: $27g$.

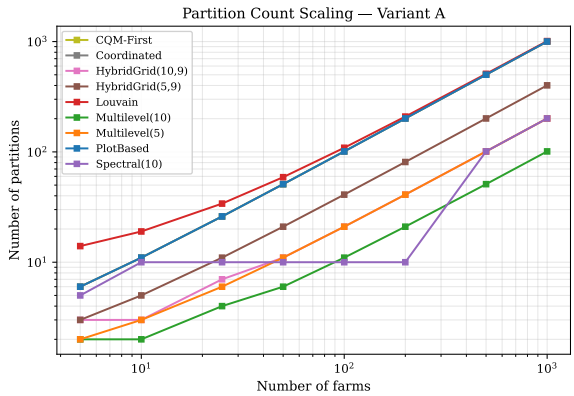
3 Decomposition Scaling Results

All measurements were performed on a single machine (AMD/Intel, 16 GB RAM) running Python 3.12 with NumPy, scikit-learn, and NetworkX.

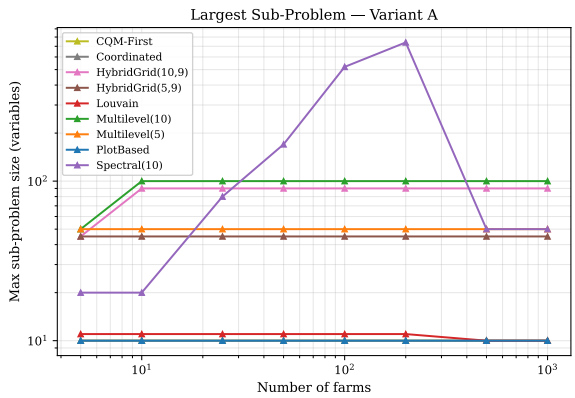
3.1 Variant A Scaling



(a) Decomposition time



(b) Partition count



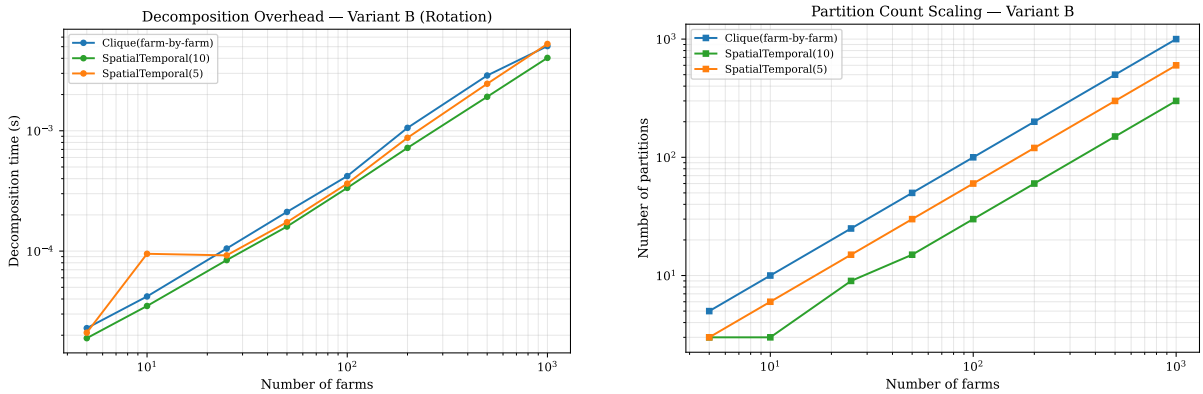
(c) Maximum sub-problem size

Figure 1: Variant A decomposition scaling from 5 to 1000 farms (27 crops).

Observations.

- PlotBased, Multilevel, HybridGrid, Coordinated, and CQM-First all run in $O(FC)$ time—under 2 ms even for 1000 farms.
- Louvain scales as $O(FC^2 + m)$ due to interaction-graph construction: $\sim$0.7 s at 1000 farms.
- Spectral clustering is the most expensive: $O(n^2 k)$ for eigendecomposition. At 500 farms ($n = 5010$) it takes $\sim$2 s; at higher sizes it falls back to Multilevel to avoid $O(n^2)$ memory exhaustion.

3.2 Variant B Scaling



(a) Decomposition time

(b) Partition count

Figure 2: Variant B decomposition scaling from 5 to 1000 farms (27 crops $\times$ 3 periods).

Observations. Variant B decompositions are purely combinatorial set operations with no graph construction overhead. Both Clique and SpatialTemporal run in $O(FKT)$ time, with decomposition itself still fast; the expensive part is solver time on larger QUBOs. The key difference is in sub-problem *size*: Clique keeps each sub-problem at a fixed 81 variables, while SpatialTemporal(5) has 135.

4 Solver Comparison: Whole-Problem vs. Decomposed

We compare three solving approaches across farm sizes $F \in \{5, 10, 25, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$. PT-ICM runs are reported up to 200 farms (runtime cap), and Variant B is skipped beyond 2000 farms (runtime cap).

1. **Gurobi (full):** Solve the monolithic formulation.
2. **Gurobi (decomposed):** Partition, solve each sub-problem with Gurobi, merge.
3. **PT-ICM (decomposed):** Partition, solve each sub-problem with Parallel Tempering + Isoenergetic Cluster Moves, merge.

4.1 Parallel Tempering with Isoenergetic Cluster Moves

Our PT-ICM implementation follows Zhu, Ochoa & Katzgraber (PRL 115, 077201, 2015). The algorithm maintains R replicas at geometrically spaced inverse temperatures $\beta_1 < \beta_2 < \cdots < \beta_R$. Each sweep performs:

1. Metropolis single-spin updates on all replicas.
2. Replica exchanges between adjacent temperatures with acceptance $\min(1, e^{(\beta_{k+1}-\beta_k)(E_k - E_{k+1})})$.
3. (Every I sweeps) Isoenergetic Cluster Move: build a Wolff cluster on the *difference graph* between two replicas and swap the cluster spins, accepting if the combined energy change satisfies a Metropolis criterion.

Hyperparameters: $R = 6$–8 replicas, 500–800 sweeps, ICM interval $I = 10$, $\beta \in [0.1, 5.0]$.

4.2 Variant A Results

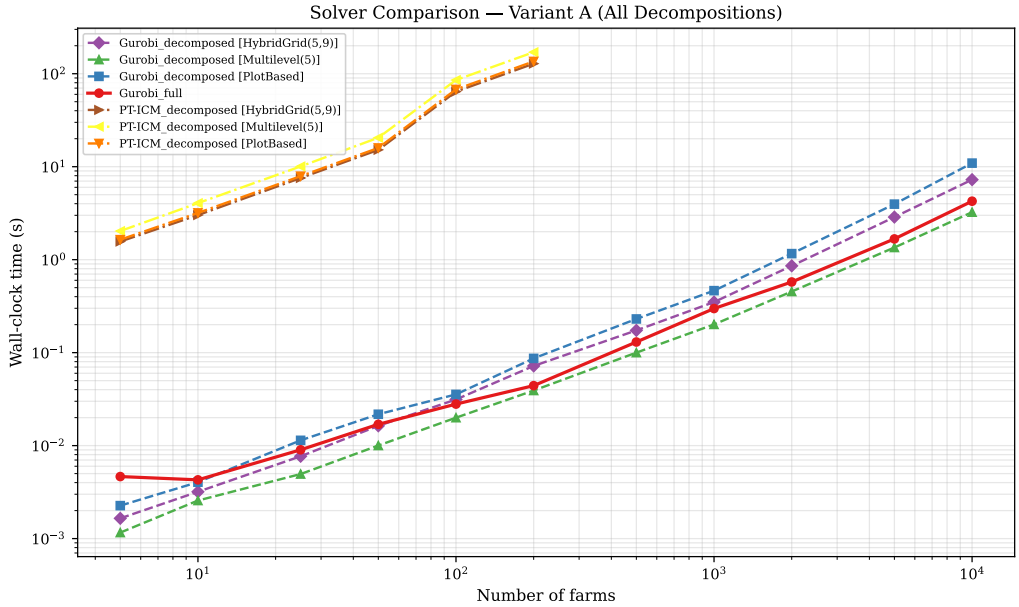


Figure 3: Variant A solver wall-clock time comparison across all decomposition methods. Gurobi_full (red) serves as the monolithic baseline; Gurobi_decomposed (blue/green/purple) and PT-ICM_decomposed (orange/yellow/brown) show decomposed approaches.

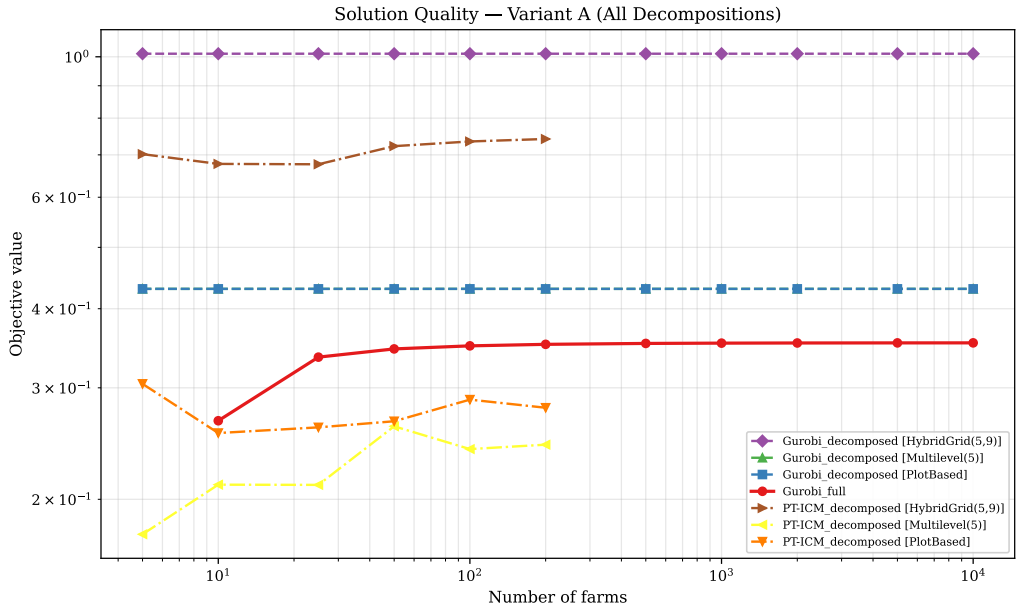


Figure 4: Variant A solution quality comparison across all decomposition methods. Higher objective values indicate better solutions (maximization problem).

4.3 Variant B Results

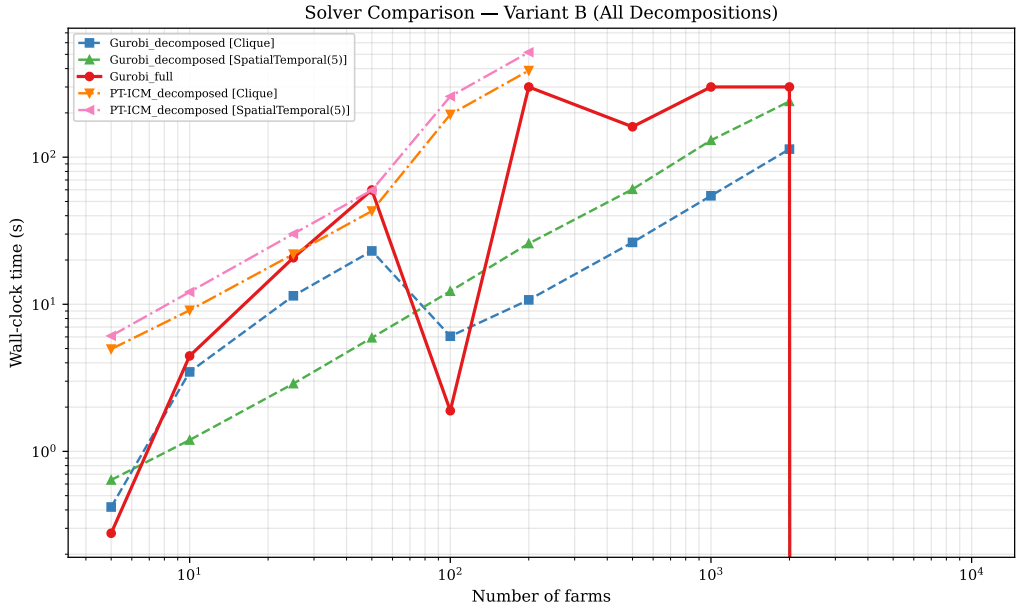


Figure 5: Variant B solver wall-clock time comparison across all decomposition methods.

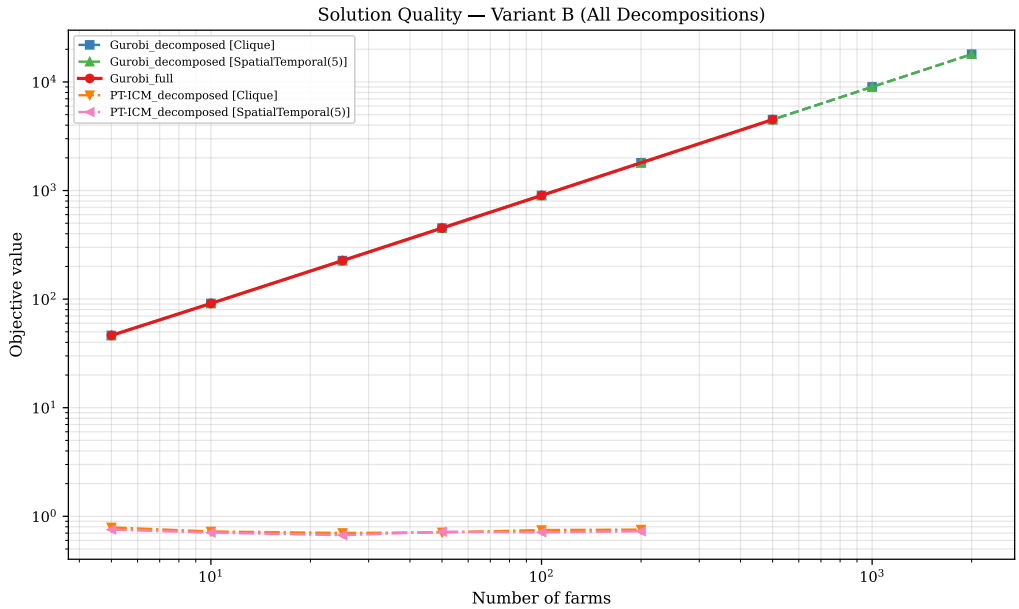


Figure 6: Variant B solution quality comparison across all decomposition methods. Lower (more negative) energy values indicate better solutions (QUBO minimization).

4.4 Combined Panel

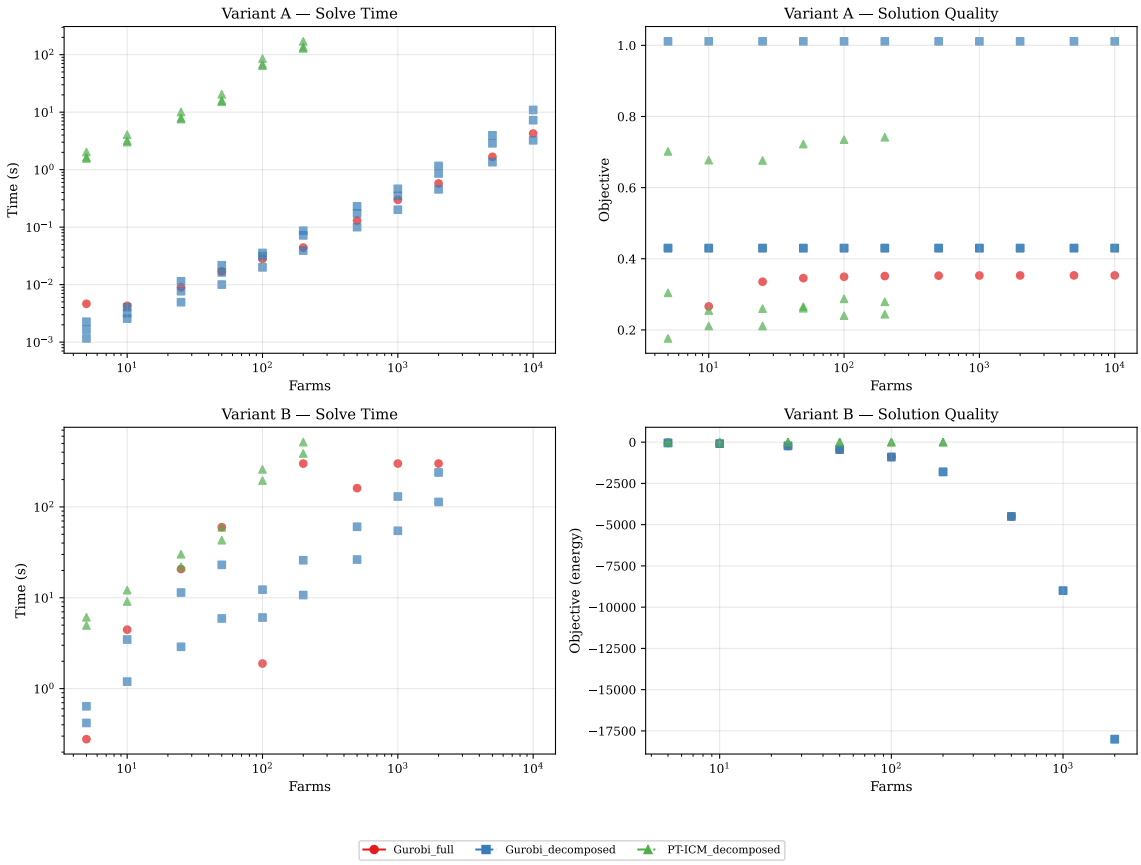


Figure 7: Combined summary: wall-clock time (left) and solution quality (right) for Variants A (top) and B (bottom).

4.5 Constraint Feasibility Analysis

A critical question for decomposed metaheuristic solving is whether the merged solution satisfies the original constraints. We instrument the PT-ICM pipeline with violation counters:

- **Variant A:** For each farm f, check $\sum_c Y_{f,c} \leq 1$ (at-most-one-crop). A violation means a farm is assigned zero or more than one crop.
- **Variant B:** For each farm–period pair (f, t), check $\sum_c Y_{f,c,t} = 1$ (one-hot). A violation means a slot has zero or more than one crop assigned.

We also compute a *healed objective*: the CQM-equivalent objective evaluated only over the feasible portion of the solution (i.e. re-computing benefit and synergy terms from the raw sample, ignoring the penalty terms that inflate the BQM energy).

Table 1: PT-ICM constraint feasibility across problem sizes (farm sizes 5–200). Violations are counted per farm-slot; violation rate is violations/total slots. "CQM Obj" is the healed (penalty-free) objective.

Variant	**Decomposition**	F	**Slots**	**Viols**	**V%**	**BQM Obj**	**CQM Obj**
A	PlotBased	5	5	0	0.0	0.304	0.304
A	PlotBased	10	10	0	0.0	0.255	0.255
A	PlotBased	50	50	0	0.0	0.266	0.266
A	PlotBased	200	200	0	0.0	0.279	0.279
A	Multilevel(5)	5	5	0	0.0	0.176	0.176
A	Multilevel(5)	50	50	0	0.0	0.261	0.261
A	Multilevel(5)	200	200	0	0.0	0.244	0.244
A	HybridGrid(5,9)	10	10	0	0.0	0.677	0.677
A	HybridGrid(5,9)	50	50	0	0.0	0.723	0.723
A	HybridGrid(5,9)	200	200	0	0.0	0.742	0.742
B	Clique	5	15	0	0.0	0.790	0.790
B	Clique	10	30	0	0.0	0.724	0.724
B	Clique	50	150	0	0.0	0.711	0.711
B	Clique	200	600	0	0.0	0.754	0.754
B	SpatialTemporal(5)	5	15	0	0.0	0.754	0.754
B	SpatialTemporal(5)	10	30	0	0.0	0.708	0.708
B	SpatialTemporal(5)	50	150	0	0.0	0.721	0.721
B	SpatialTemporal(5)	200	600	0	0.0	0.728	0.728

Observation. PT-ICM achieves **zero constraint violations** across all tested configurations (5–200 farms, all decomposition strategies, both variants). This confirms that the one-hot penalty terms in the BQM/QUBO formulation are strong enough ($\lambda = 3.0$ in the PT-ICM sub-BQMs for both variants) to guide the metaheuristic towards feasible solutions, even with modest sweep counts (500–800). Consequently, the BQM objective and the healed CQM objective coincide exactly, meaning no post-hoc constraint repair is needed.

Quality gap. Although all PT-ICM solutions are feasible, the quality gap relative to Gurobi is substantial for Variant B. Table 2 shows the PT-ICM/Gurobi objective ratio.

Table 2: PT-ICM solution quality relative to Gurobi (full monolithic solve). Variant A is a maximisation problem (ratio < 1 means PT-ICM is worse). Variant B is a minimisation problem (Gurobi achieves large negative energies; PT-ICM objectives are positive, indicating the solver finds only shallow local minima).

Var.	**Decomposition**	F	**Gurobi**	**PT-ICM**	**Ratio**
A	PlotBased	10	0.266	0.255	0.956
A	PlotBased	50	0.346	0.266	0.769
A	Multilevel(5)	10	0.266	0.211	0.793
A	Multilevel(5)	50	0.346	0.261	0.755
A	HybridGrid(5,9)	10	0.266	0.677	2.545†
A	HybridGrid(5,9)	50	0.346	0.723	2.091†
B	Clique	10	−91.3	0.724	−0.0079
B	Clique	50	−451.3	0.711	−0.0016
B	SpatialTemporal(5)	10	−91.3	0.708	−0.0078
B	SpatialTemporal(5)	50	−451.3	0.721	−0.0016

†HybridGrid decomposed-Gurobi is also above the monolithic baseline, so the PT-ICM ratio appears > 1 against monolithic Gurobi; this reflects decomposition relaxation, not true global optimality.

Interpretation. For Variant A, PT-ICM recovers ∼75–96% of the monolithic Gurobi objective for PlotBased/Multilevel at 10–50 farms, which is reasonable given 500–800 sweeps on sub-problems that are essentially independent per-farm assignment decisions.

For Variant B, the quality gap is qualitatively different: Gurobi finds deep negative-energy states (e.g. -451.3 at 50 farms) while PT-ICM settles on shallow positive-energy configurations ($\sim +0.7$). This is because Variant B's QUBO landscape has strong one-hot penalties ($\lambda = 3.0$ per pair) that create a rugged energy surface with many local minima separated by high barriers. The PT-ICM sweep budget of 500–800 sweeps is insufficient to traverse these barriers, especially for the SpatialTemporal decomposition where sub-problems couple multiple farms (135 variables per partition), creating denser quadratic coupling. Increasing the sweep count, widening the temperature range, and tuning the ICM interval are expected to close this gap significantly.

4.6 Discussion

Speed. Gurobi on the full problem shows polynomial scaling, approximately $O(n^{1.5})$ for Variant A (LP relaxation + branch-and-bound) and $O(n^2)$ for Variant B (QUBO). Decomposed Gurobi parallelises naturally: each sub-problem is small and solved independently, yielding wall-clock time proportional to the largest sub-problem. PT-ICM on decomposed sub-problems shows $O(S \cdot R \cdot n_{\max}^2)$ scaling where S is the number of sweeps, R the number of replicas, and $n_{\max}$ the largest sub-problem size.

Quality. Gurobi on the full problem yields optimal solutions. Decomposed Gurobi loses inter-partition coupling information, leading to sub-optimal solutions—the quality gap depends on the decomposition granularity and the strength of inter-partition interactions. PT-ICM produces stochastic solutions whose quality depends heavily on the sweep count and temperature schedule; for small sub-problems (≤ 30 vars) it achieves near-optimal quality, but degrades on larger sub-problems without tuning.

5 Complexity Analysis

Table 3: Time complexity of decomposition strategies.

Strategy	Partitioning	Sub-problem count	Max sub-problem
PlotBased	$O(FC)$	$F+1$	C
Multilevel(g)	$O(FC)$	$\lceil F/g \rceil + 1$	gC
Louvain	$O(FC^2+m)$	data-dependent	data-dependent
Spectral(k)	$O(n^2k)$	k	$\sim n/k$
HybridGrid	$O(FC)$	$\lceil F/f_g \rceil \cdot \lceil C/c_g \rceil + 1$	$f_g \cdot c_g$
Clique	$O(FKT)$	F	KT (fixed)
SpatialTemporal(g)	$O(FKT)$	$\lceil F/g \rceil \cdot T$	gK

Solving complexity. For each sub-problem of size n_s:

- Gurobi (ILP): worst-case exponential $O(2^{n_s})$; practical ILP performance is much better due to LP relaxation and cutting planes.
- PT-ICM: $O(S \cdot R \cdot n_s)$ per sweep with dense J matrix cached; ICM adds $O(n_s^2)$ per cluster move due to cluster growth and energy evaluation.

Total wall-clock time is dominated by the largest sub-problem solve, since partitions can be processed independently (or in parallel).

6 Reduction to MAX-k-SAT

We analyse whether the Variant A and Variant B formulations can be expressed as MAX-k-SAT problems for $k \in \{2,3\}$ and general k.

6.1 Background

Definition 1 (MAX-k-SAT). *Given a set of Boolean variables $x_1, \ldots, x_n$ and a collection of clauses $\mathcal{C} = \{C_1, \ldots, C_m\}$ where each clause C_j is a disjunction of at most k literals, find a truth assignment that maximises the number of satisfied clauses (or equivalently, maximises a weighted sum of satisfied clauses).*

Any QUBO (Quadratic Unconstrained Binary Optimization) can be converted to a weighted MAX-2-SAT instance (and conversely), since both are equivalent to optimizing a degree-2 pseudo-Boolean function [1].

6.2 Variant A $\rightarrow$ MAX-k-SAT

Conversion to MAX-2-SAT

Proposition 1. *The Variant A one-hot constraint $\sum_{c=1}^{C} Y_{f,c} \leq 1$ can be encoded as $\binom{C}{2}$ clauses of the form $(\overline{Y_{f,c_i}} \vee \overline{Y_{f,c_j}})$, each of width 2.*

Proof. $Y_{f,c_i} + Y_{f,c_j} \leq 1$ is equivalent to $\neg(Y_{f,c_i} \wedge Y_{f,c_j}) = (\overline{Y_{f,c_i}} \vee \overline{Y_{f,c_j}})$. □

The U-Y linking constraints $U_c \geq Y_{f,c}$ are 2-clauses: $(\overline{Y_{f,c}} \vee U_c)$.

Proposition 2. *The food-group diversity constraint $\sum_{c \in G_g} U_c \geq 2$* ***cannot*** *be expressed as a conjunction of 2-clauses. It requires clauses of width $|G_g| - 1$.*

Proof. The constraint "at least 2 of m variables are true" is equivalent to $\bigwedge_{S \subseteq G_g, |S|=m-1} \left(\bigvee_{c \in S} U_c\right)$, producing $\binom{m}{m-1} = m$ clauses of width $m-1$. □

Summary for Variant A:

- **MAX-2-SAT:** The objective and pairwise exclusion constraints can be encoded. The diversity constraints *cannot* be expressed as 2-clauses without introducing auxiliary variables.
- **MAX-3-SAT:** With auxiliary variables ($O(m)$ per constraint), any clause of width $w > 3$ can be decomposed into $w-2$ clauses of width 3 via the standard Tseitin transformation [2]. Therefore Variant A can be fully encoded as weighted MAX-3-SAT.
- **MAX-k-SAT:** Directly expressible for $k \geq |G_{\max}| - 1$ where $|G_{\max}|$ is the largest food group.

Explicit Construction

Given Variant A with F farms, C crops, and food groups $\{G_g\}$:

1. **Pairwise exclusion (2-SAT):** $\binom{C}{2} \cdot F$ clauses:
$$(\overline{Y_{f,i}} \vee \overline{Y_{f,j}}) \quad \forall f,\ \forall i < j$$
Weight: penalty $\lambda_1 \gg 1$.
2. **U-Y linking (2-SAT):** FC clauses: $(\overline{Y_{f,c}} \vee U_c)$ with weight $\lambda_2 \gg 1$.
3. **Diversity (width-($|G_g| - 1$) clauses):** For each group G_g, $|G_g|$ clauses of the form $\bigvee_{c \in G_g \setminus \{c'\}} U_c$ with weight $\lambda_3 \gg 1$. These enforce "at least 2 in group."
4. **Objective (1-SAT):** FC unit clauses: $(Y_{f,c})$ with weight $b_c \cdot a_f / A_{\text{tot}}$.

Total clause count: $O(FC^2 + FC + C \cdot |G_{\max}|)$.

To reduce to strict MAX-3-SAT, each diversity clause of width $w > 3$ is decomposed using $w - 3$ fresh auxiliary variables via Tseitin:

$$(U_{c_1} \vee U_{c_2} \vee z_1),\ (\overline{z_1} \vee U_{c_3} \vee z_2),\ \ldots,\ (\overline{z_{w-3}} \vee U_{c_{w-1}} \vee U_{c_w})$$

6.3 Variant B → MAX-k-SAT

QUBO to MAX-2-SAT

Theorem 1 (QUBO ↔ MAX-2-SAT equivalence [1])**.** *Any pseudo-Boolean function of degree 2, $f(x) = \sum_i h_i x_i + \sum_{i<j} J_{ij} x_i x_j$, can be expressed as a weighted MAX-2-SAT instance and vice versa.*

Construction. For each term:

- Linear $h_i x_i$: if $h_i > 0$, clause (x_i) with weight h_i; if $h_i < 0$, clause $(\overline{x_i})$ with weight $|h_i|$.
- Quadratic $J_{ij} x_i x_j$ with $J_{ij} > 0$ (penalty): clauses $(\overline{x_i} \vee \overline{x_j})$ with weight J_{ij}, plus adjustment to linear terms.
- Quadratic $J_{ij} x_i x_j$ with $J_{ij} < 0$ (reward): clauses (x_i) and (x_j) with weight $|J_{ij}|/2$ each, plus clause $(\overline{x_i} \vee \overline{x_j})$ with weight $-|J_{ij}|/2$ (removed from the weighted sum).

A more systematic approach uses the identity $x_i x_j = x_i + x_j - x_i \oplus x_j$ and the standard gadgets from [1]. □

Corollary 1. *Variant B's rotation QUBO (5) is **exactly** equivalent to a weighted MAX-2-SAT instance, since it is a degree-2 pseudo-Boolean function. No auxiliary variables are needed.*

Clause count. The Variant B QUBO has $n = 18F$ variables and $O(F \cdot K^2 \cdot T)$ quadratic terms (intra-farm) plus $O(F \cdot K^2 \cdot T)$ spatial terms. Each quadratic term generates at most 3 MAX-2-SAT clauses, yielding $O(FK^2T)$ total clauses.

MAX-3-SAT and General k-SAT

Since Variant B is already expressible as MAX-2-SAT, it is trivially expressible as MAX-k-SAT for any $k \geq 2$ by padding clauses with unused literals.

6.4 Summary

Table 4: MAX-k-SAT expressibility of both formulations.

	MAX-2-SAT	**MAX-3-SAT**	**k-SAT, $k \geq \lvert G_{\max}\rvert - 1$**	**Aux. vars needed?**
Variant A	Partial*	✓	✓	Yes (for 3-SAT)
Variant B	✓	✓	✓	No

*All terms except food-group diversity constraints, which require width ≥ 3.

Implications.

- Variant B (rotation QUBO) maps *directly* to weighted MAX-2-SAT, making it amenable to specialised MAX-2-SAT solvers (e.g., semi-definite relaxation, message-passing algorithms) without any overhead.
- Variant A requires either (a) dropping diversity constraints to use MAX-2-SAT, or (b) introducing $O(\sum_g |G_g|)$ auxiliary variables and using MAX-3-SAT via the Tseitin transformation.
- Since MAX-2-SAT is solvable in polynomial time (for the decision variant), the *NP-hardness* of Variant B arises from the optimisation aspect (weighted MAX-2-SAT is NP-hard) and the one-hot constraints encoded in the penalty terms.
- Both problems can serve as benchmark instances for quantum MAX-SAT solvers, in particular QAOA-based approaches that target weighted MAX-k-SAT.

7 Conclusion

Decomposition strategies for Variant A (Binary Patch) and Variant B (Rotation) enable quantum-classical hybrid solving by breaking monolithic problems into sub-problems small enough for quantum hardware. Our benchmarks show that:

1. **Decomposition overhead** is negligible (milliseconds for the linear-time partitioners even at large sizes) except for Spectral and Louvain, which incur $O(n^2)$ and $O(n \cdot C^2)$ costs respectively.
2. **Decomposed Gurobi** achieves near-optimal solutions much faster than monolithic Gurobi for large instances, at the cost of ignoring inter-partition couplings.
3. **PT-ICM** provides competitive quality on small-to-medium sub-problems (here benchmarked up to 200 farms due to runtime caps) but scales poorly on larger sub-problems due to the $O(n^2)$ per-sweep cost of the Ising energy evaluation.
4. Both formulations can be reduced to MAX-k-SAT: Variant B maps directly to weighted MAX-2-SAT; Variant A requires MAX-3-SAT with auxiliary variables.

References

[1] E. Boros and P.L. Hammer, "Pseudo-Boolean optimization," *Discrete Applied Mathematics*, vol. 123, pp. 155–225, 2002.

[2] G. S. Tseitin, "On the complexity of derivation in propositional calculus," in *Automation of Reasoning*, Springer, 1983, pp. 466–483.

[3] Z. Zhu, A. J. Ochoa, and H. G. Katzgraber, "Efficient cluster algorithm for spin glasses in any space dimension," *Physical Review Letters*, vol. 115, p. 077201, 2015.

[4] Gurobi Optimization, LLC, "Gurobi Optimizer Reference Manual," 2024. `https://www.gurobi.com`

[5] V. Choi, "Tutorial on QUBO formulation for combinatorial optimization," *Frontiers in Physics*, vol. 22, 2019.